Finance Fraud & Loan Risk Analysis

Case Study

Financial institutions process thousands of transactions daily across customers, accounts, and branches. Understanding these financial activities is essential for identifying fraud risks, monitoring customer behavior, and evaluating loan exposure. However, large banking datasets are often complex and distributed across multiple tables, making manual analysis inefficient.

This project analyzes banking transaction and loan data to uncover patterns in financial activity. The data was cleaned and prepared using Python, and interactive dashboards were developed using Tableau to explore transaction behavior, customer demographics, loan patterns, and branch performance.

Business Objective

- Identify patterns in banking transactions.
- Understand customer demographics and financial behavior.
- Analyze transaction types and monthly financial activity.
- Monitor loan distribution and potential risk exposure.
- Evaluate branch-level performance across the banking network.
- Provide insights to support fraud monitoring and financial decision-making.

Dataset Information

- Data set source: Kaggle
- Dataset:Finance Fraud and Loan Datadet
- Records: 49,565 transactions
- Customers: 862
- Accounts: 1,651

Key attributes include transaction amount, transaction type, branch name, loan amount, loan status, interest rate, and customer demographic details.

Tools & Technologies Used

- Python – Data preprocessing and cleaning
- Pandas – Data manipulation
- NumPy – Numerical computation
- Tableau – Interactive dashboards and visualization

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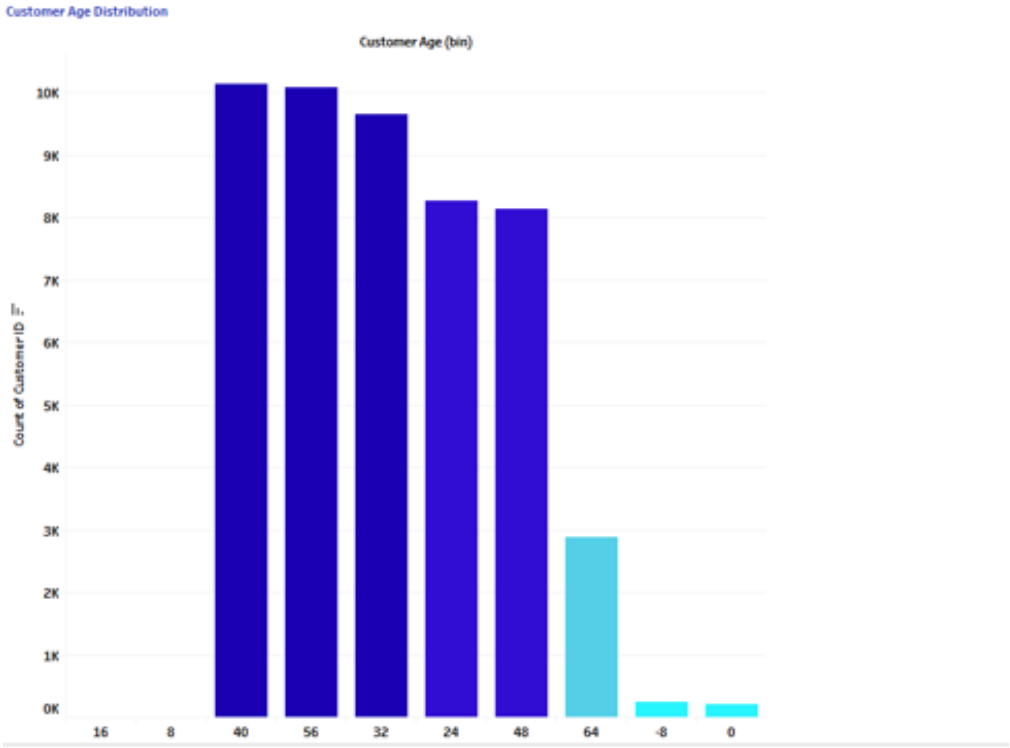
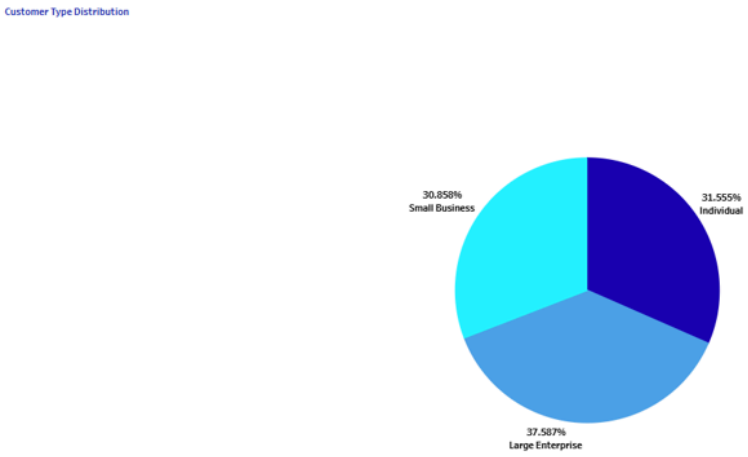
Figure 1 – Customer Age Distribution



Key Insights:

1. Most customers fall within the economically active age group (25–55).
2. Middle-aged customers show the highest participation in banking transactions.
3. Very young and elderly customers contribute less to financial activity.

Figure 2 – Customer Type Distribution



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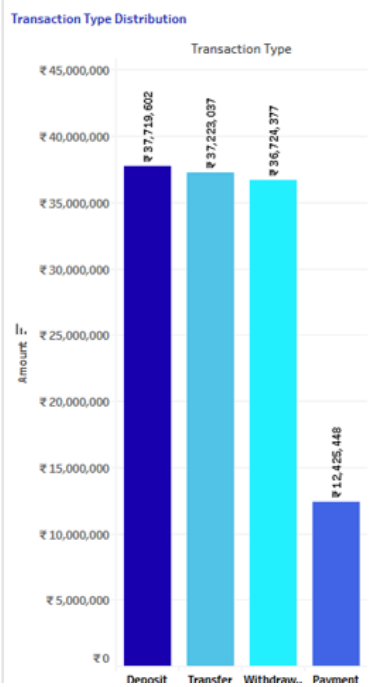
Key Insights:

1. Customer segments are distributed across Individual, Small Business, and Large Enterprise customers.
2. Large enterprise customers contribute the largest share of financial activity.
3. Small businesses represent a significant portion of transaction volume.

Figure 3 – Transaction Type Distribution



Key Insights:

1. Deposits, transfers, and withdrawals dominate transaction activity.
2. Payment transactions represent the smallest portion of financial activity.
3. Deposit transactions generate the highest financial value.

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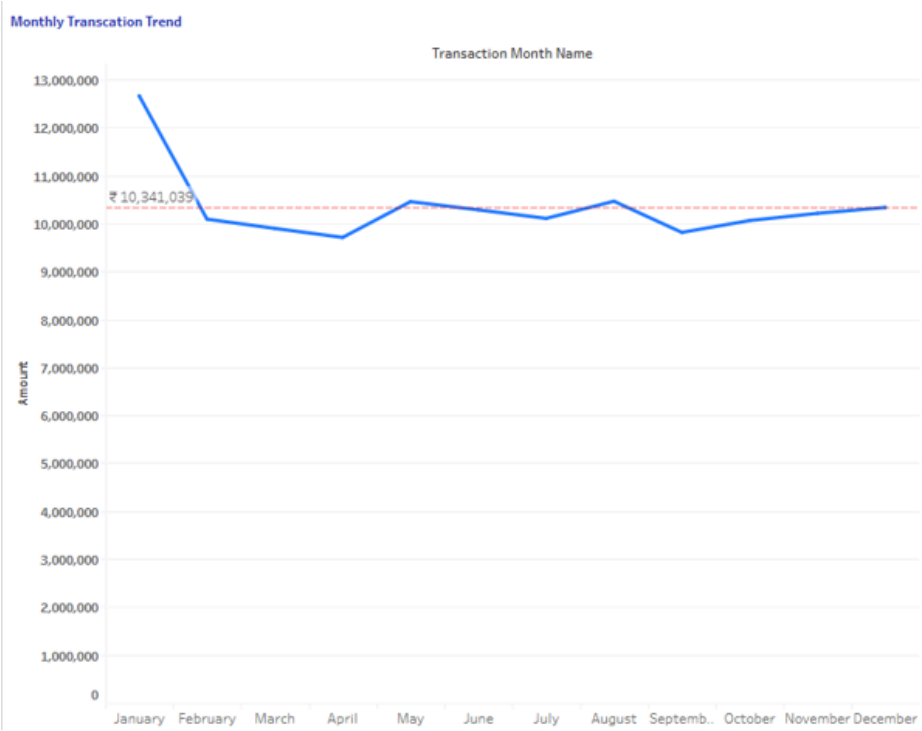
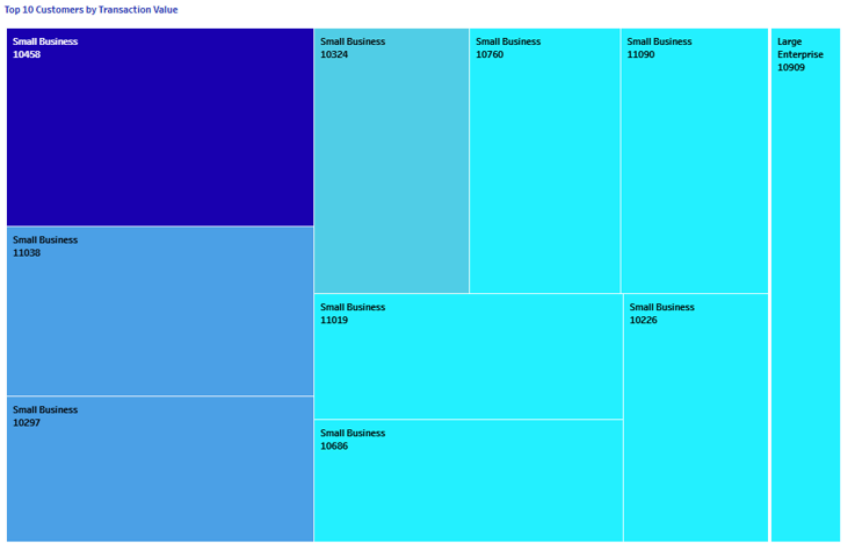
Figure 4 – Monthly Transaction Trend



Key Insights:

1.Transaction volume remains relatively stable across months.
2. January shows the highest financial activity.
3. Minor fluctuations suggest consistent banking usage.

Figure 5 – Top Customers by Transaction Value



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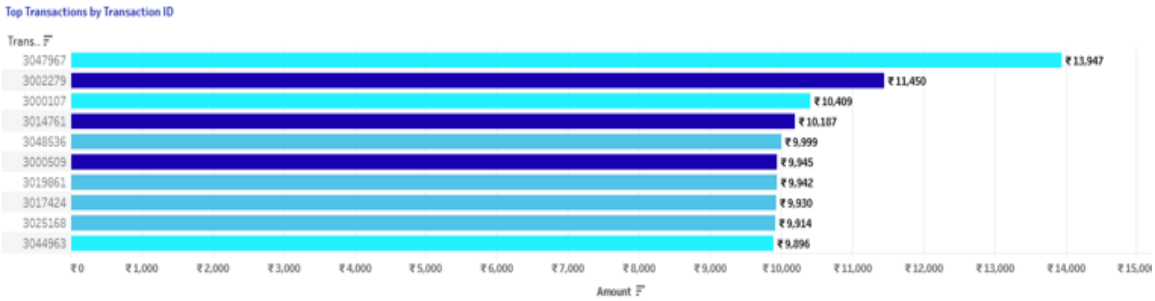
Key Insights:

1. A small number of customers contribute disproportionately to transaction volume.
2. Small businesses dominate among high transaction value customers.
3. Large enterprises also contribute significant financial activity.

Figure 6 – Top Transactions by Transaction ID



Key Insights:

1. Several transactions exceed ₹10,000 in value.
2. High-value transactions require monitoring for fraud detection.
3. Transaction activity is concentrated among a small subset of accounts.

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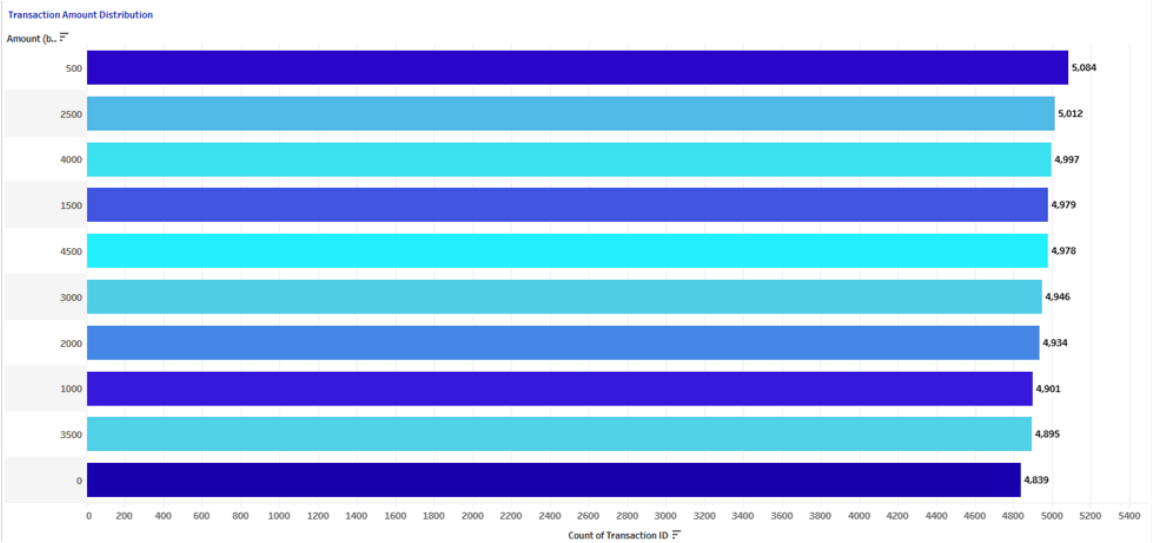
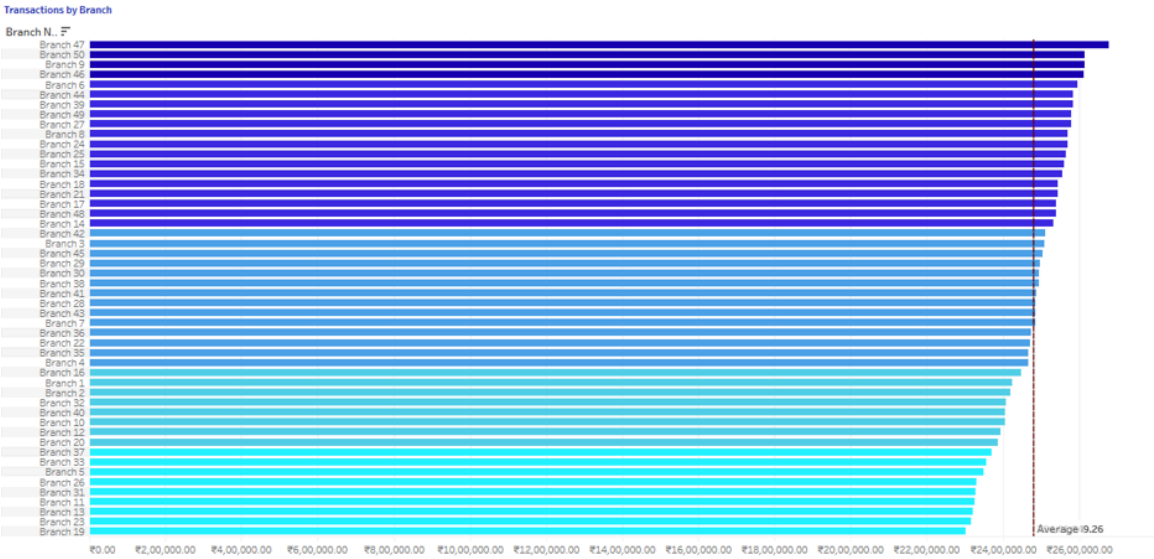
Figure 7 – Transaction Amount Distribution



Key Insights:

1. Most transactions occur within moderate value ranges.
2. Large transactions are fewer but significant in financial impact.
3. Transaction frequency peaks in mid-value ranges.

Figure 8 – Transactions by Branch

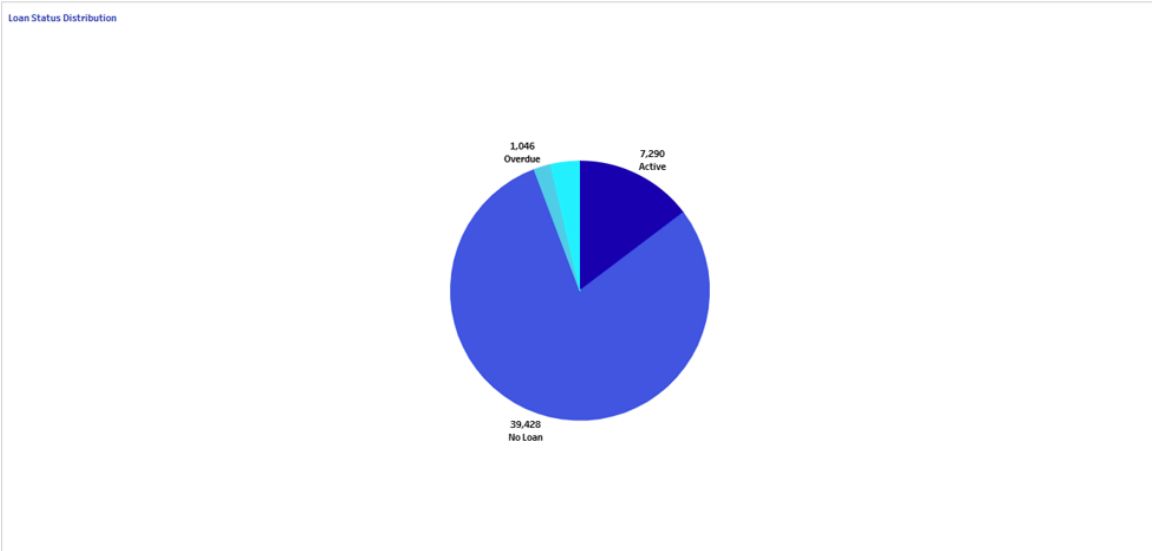


Key Insights:

1. Certain branches perform significantly above the average transaction level.
2. Branch performance varies across regions.

3. Some branches generate consistently high financial activity.

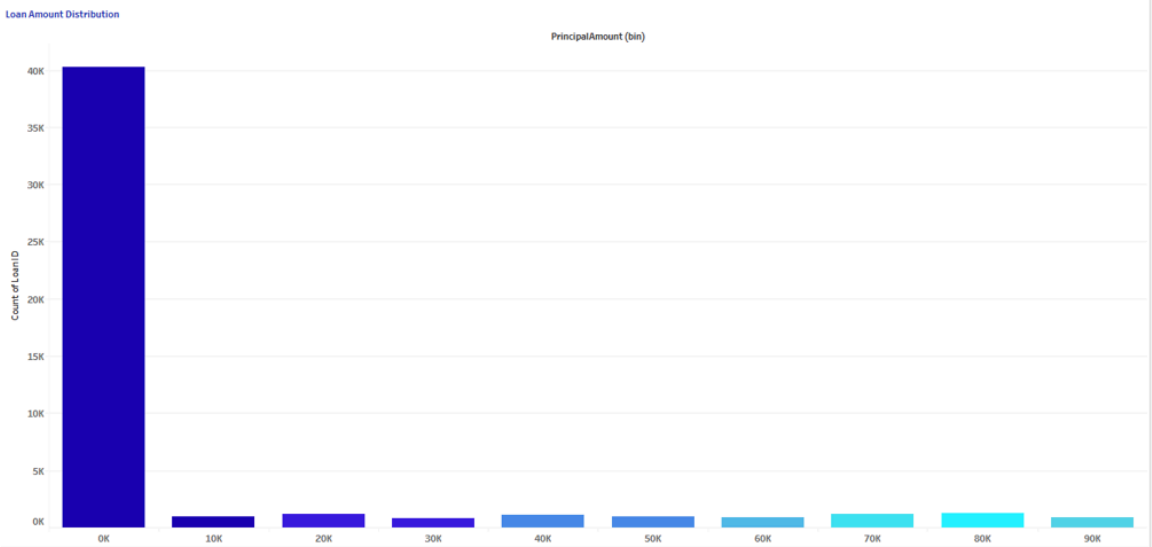
Figure 9 – Loan Status Distribution



Key Insights:

1. Most customers currently have no active loan.
2. Active loans represent a smaller but important segment.
3. Overdue loans represent potential financial risk.

Figure 10 – Loan Amount Distribution



Key Insights:

1. Most loans fall within lower principal ranges.

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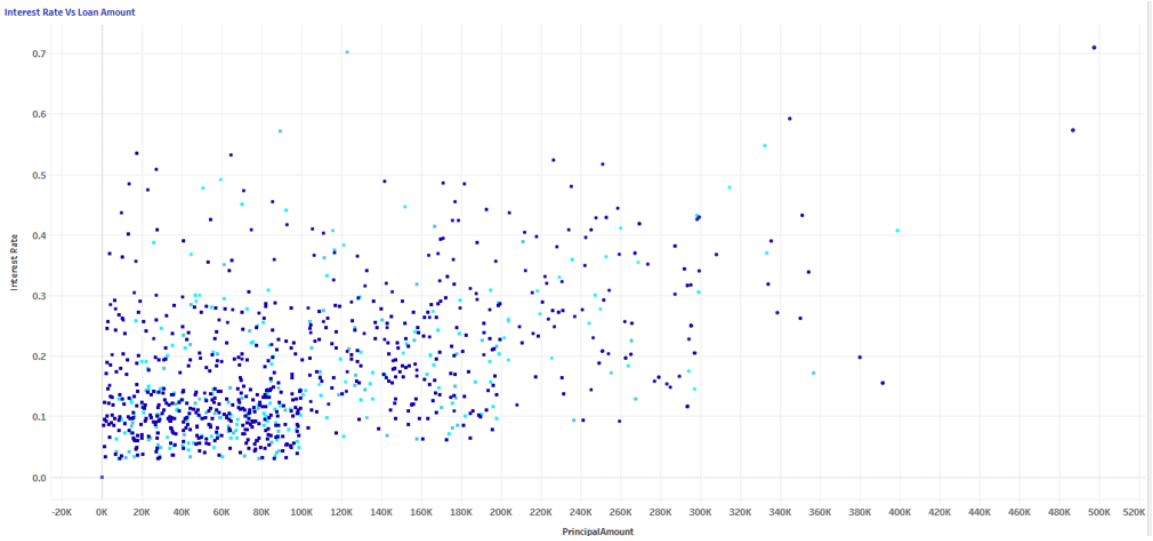
2. Higher-value loans represent smaller but financially significant exposure.
3. Loan distribution indicates varied credit risk.

Figure 11 – Interest Rate vs Loan Amount

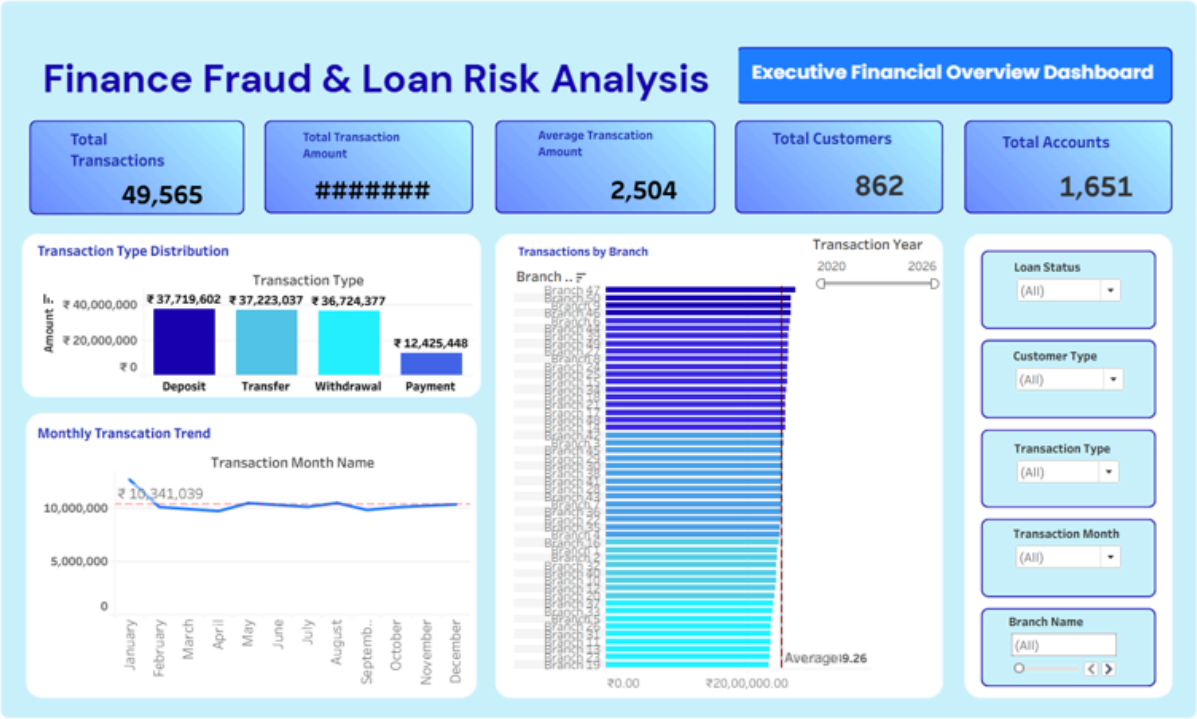


Key Insights:

1. Interest rates vary significantly across loan sizes.
2. Higher loan amounts show wider interest rate variability.
3. Loans with high principal and high interest may indicate higher risk.

Dashboard Overview

Dashboard 1 – Executive Financial Overview



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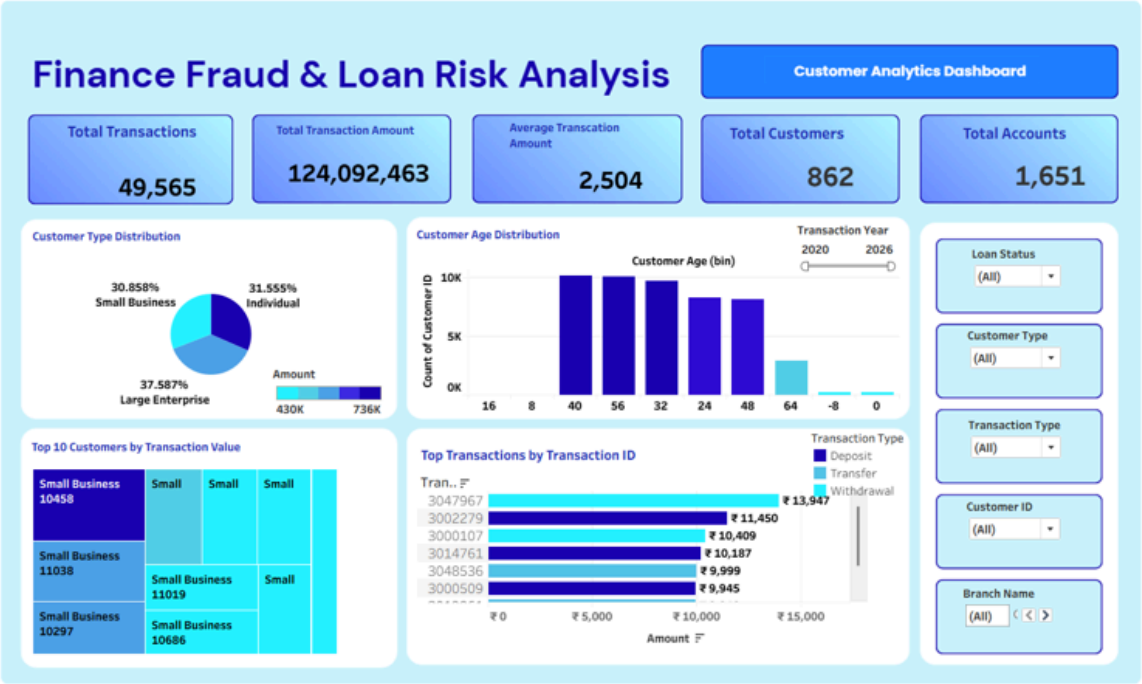
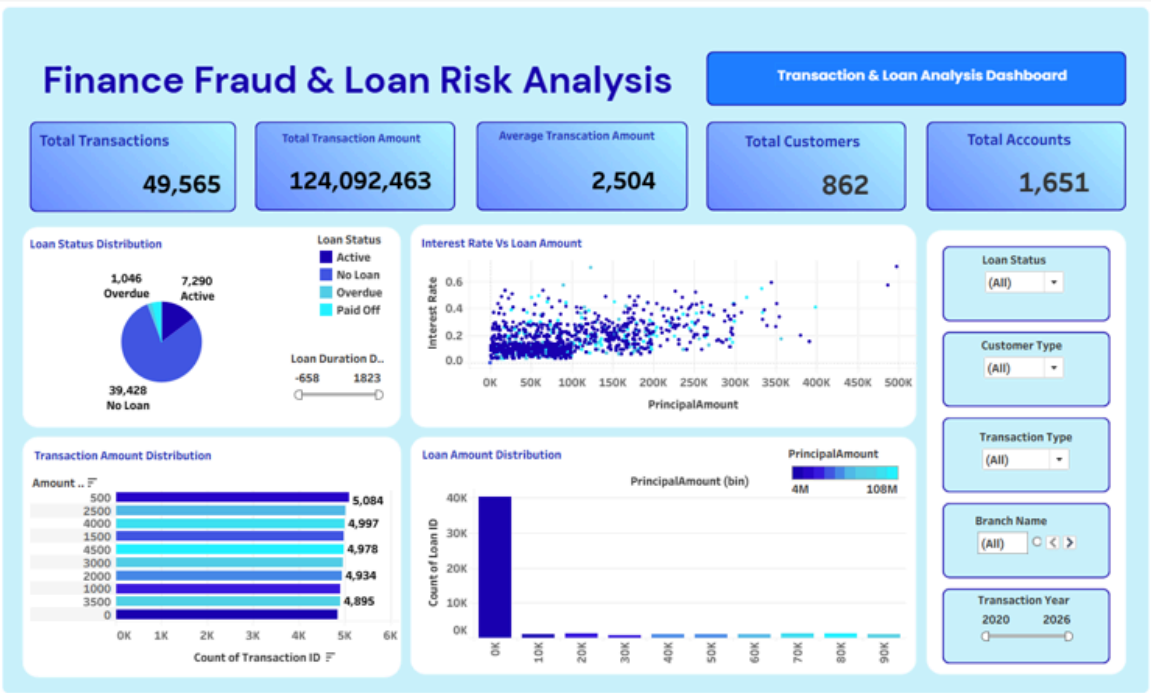
Dashboard 2 – Customer Analytics Dashboard



Dashboard 3 – Transaction & Loan Analysis Dashboard



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Project Outcome

This project demonstrates how financial transaction and loan data can be analyzed to uncover meaningful patterns in customer behavior and financial activity. Using Python for preprocessing and Tableau for visualization, the analysis reveals transaction trends, customer segmentation patterns, branch performance differences, and potential loan risk exposure. These insights can support financial institutions in improving fraud detection, monitoring financial activity, and making data-driven decisions.